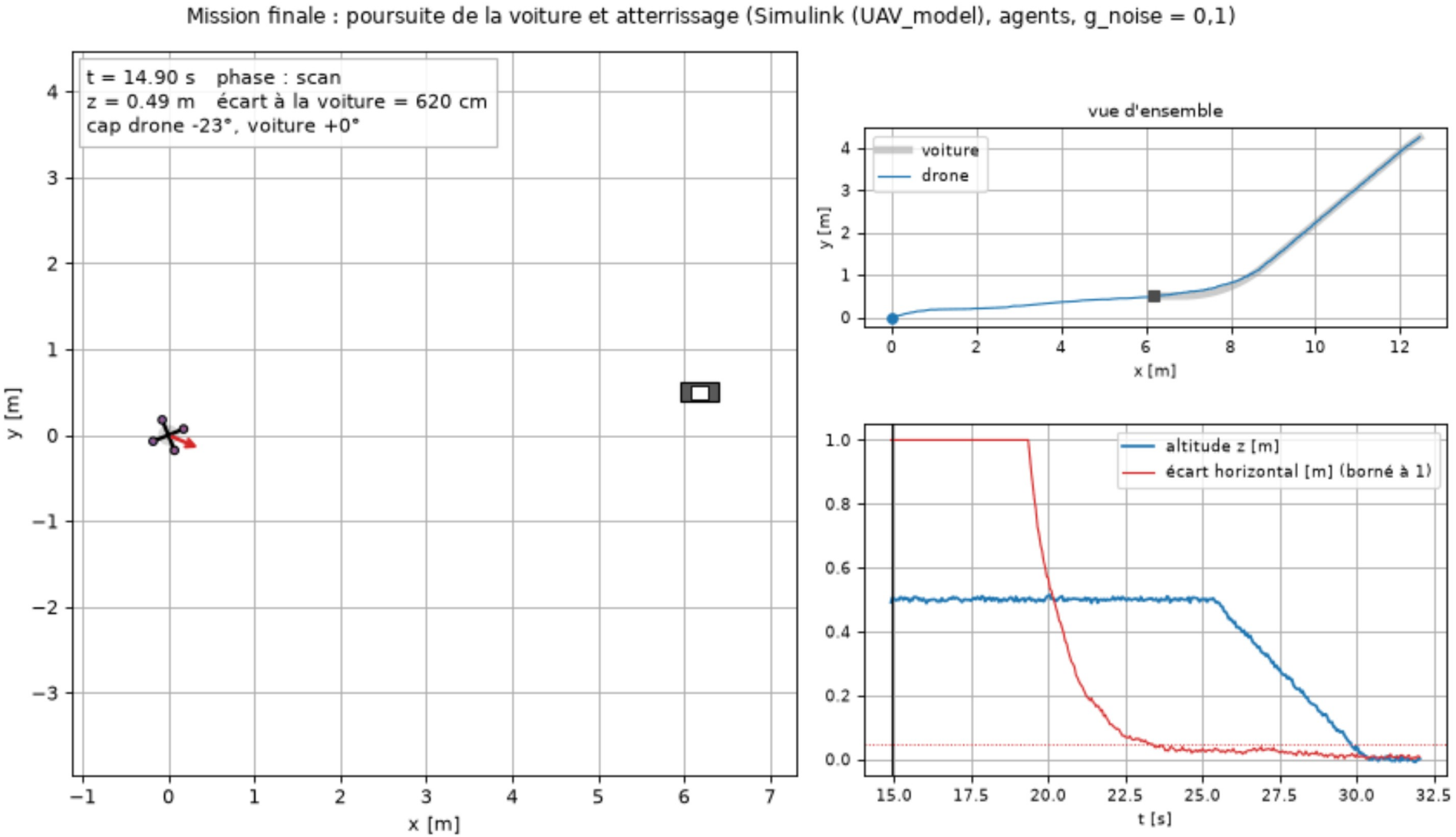
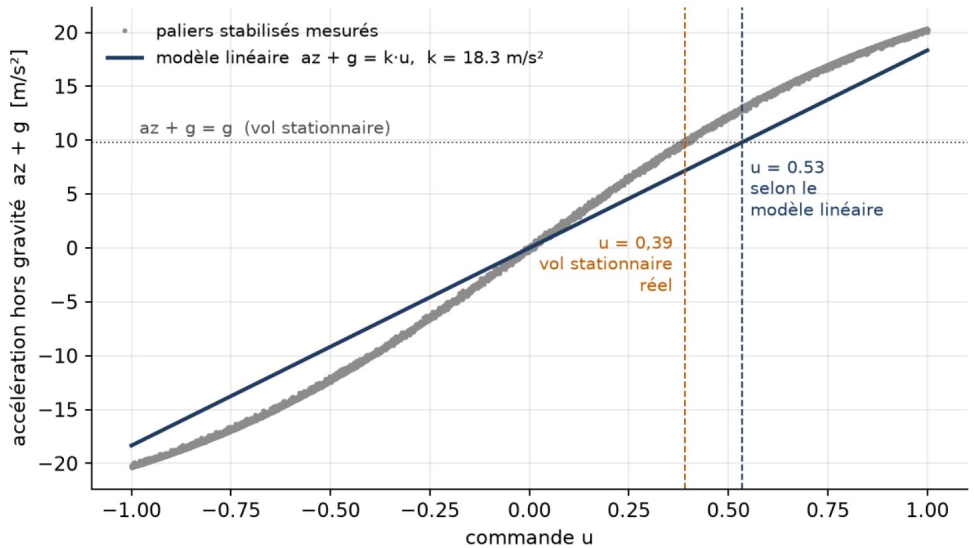
Our reinforcement-learning agents take off, scan 360°, chase a turning car and land on it.



Simulink flight, RL agents, g_noise = 0.1: landed at 30.1 s, 0.8 cm from the centre

Our reference is an LQ regulator on a double integrator, with an integral state that corrects the hover point the linear model gets wrong.



Measured static thrust (grey) vs the least-squares line $a_z + g = k \cdot u$. The line puts hover at **u = 0.53**; the real hover is **u = 0.39**.

Without the integral
plain LQR settles at **0.9 m** for a 1 m target

1 · MODEL

$$\dot{x} = A\,x + B\,u,\ x = [z,\ v_z]$$

A = [0 1; 0 0], B = [0; k], k ≈ 18.4 m/s², zero-order hold at Ts = 0.05 s

2 · LQ SYNTHESIS

$$J = \Sigma\, x^T Q x + R\,u^2 \rightarrow u = -K\,x$$

discrete Riccati equation solved by dlqr

3 · INTEGRAL STATE

$$x_a = [z,\ v_z,\ \int e]$$

removes the static error left by the wrong hover point

4 · CONTROL LAW

$$u = u_0 - K_p e - K_d v_z - K_i \int e$$

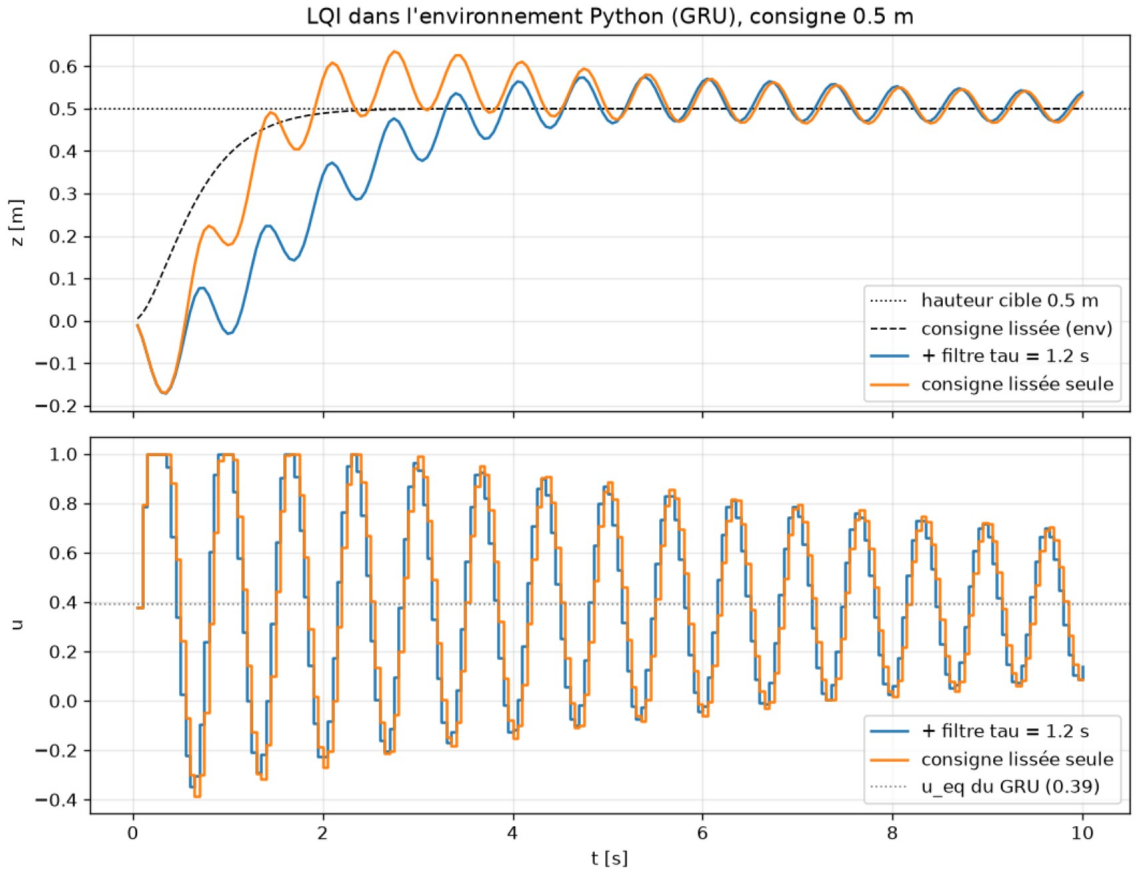
$u_0 \approx 0.39$, $|u| \le 1$, reference filter $\tau = 1.2$ s, anti-windup

Axis	Weights	Gains K_p / K_d / K_i
z	Q = diag(80, 25, 25), R = 1	**1.86 / 0.88 / 0.79** (2.32 / 1.10 / 0.98 × 0.8)
yaw ψ	k_ψ = 168 rad/s², Q = diag(120, 40, 30), R = 10	**0.252 / 0.125 / 0.097**
pitch θ	Q = diag(150, 15, 100), R = 1	**0.443 / 0.130 / 0.280**
x (PD)	$\omega_n = \sqrt{(g \cdot K_p)}$ = 1.5 rad/s, ζ = 0.9	**K_p 0.23, K_d 0.28**, $\lvert\theta_{ref}\rvert \le 0.2$ rad

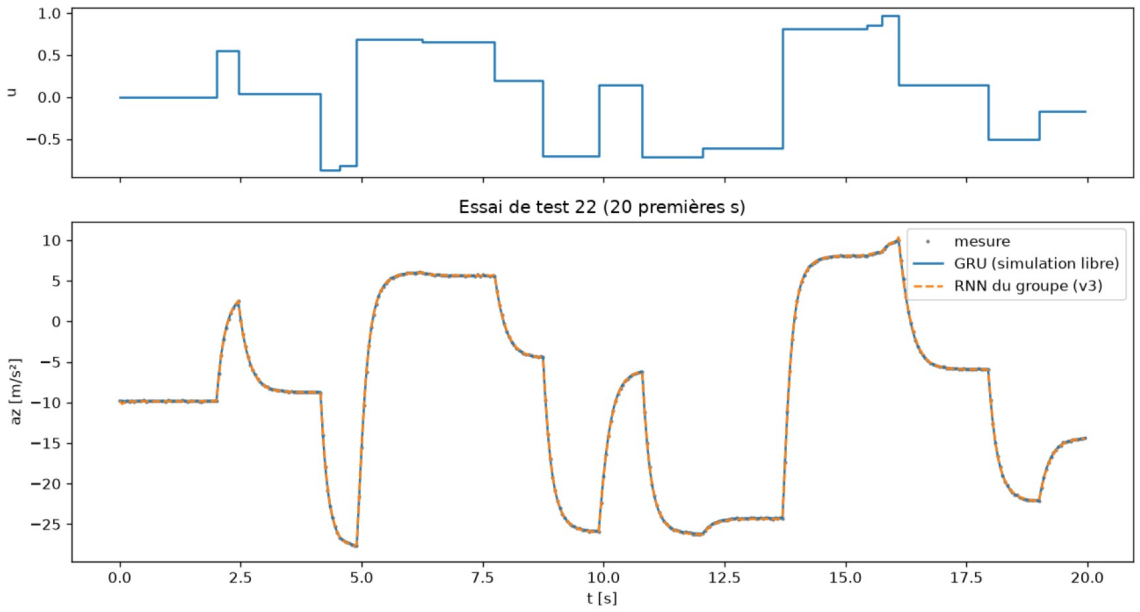
Gains then fit so that the 20 Hz response matches the continuous Simulink response (z: 5.2 → 2.6 cm; θ: diverging → 0.021 rad). “LQI” means this version from now on.

At 20 Hz the LQI built on the linear model oscillates; a GRU trained on the 2996 flights reproduces the drone down to the sensor noise.

Linear-model LQI, 0.5 m step, loop at 50 ms



GRU (1 layer, 64 units), free run on a test flight



GRU test error 0.045 m/s², at the noise floor (0.044) · it is the drone the agents train on

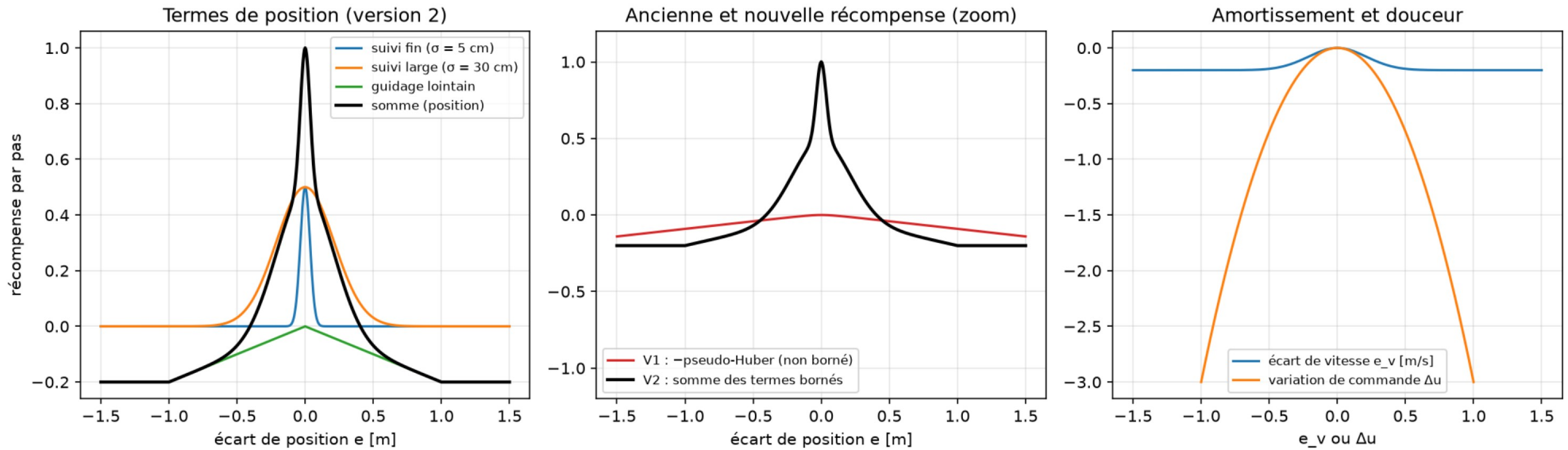
Agents learn only on models identified from data, then are checked step by step before flying on Simulink.



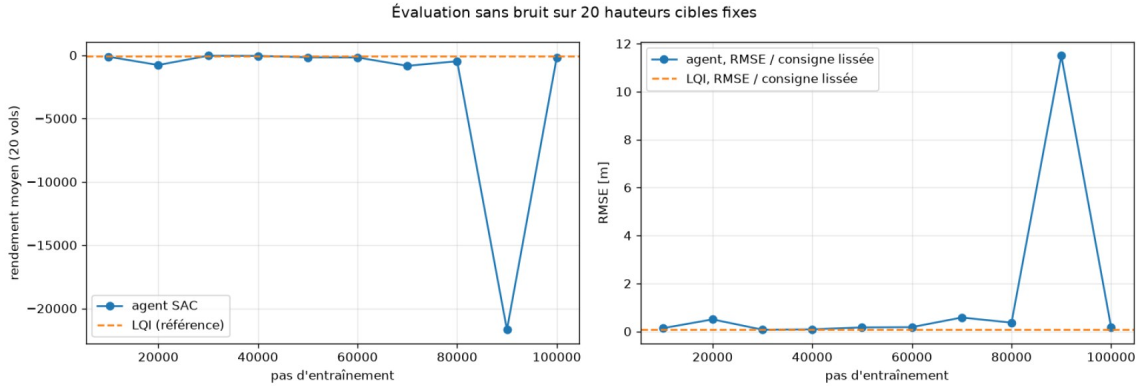
No retraining between the identified models and Simulink

A bounded reward that guides from far and rewards precision up close; PPO learns it steadily where SAC collapsed.

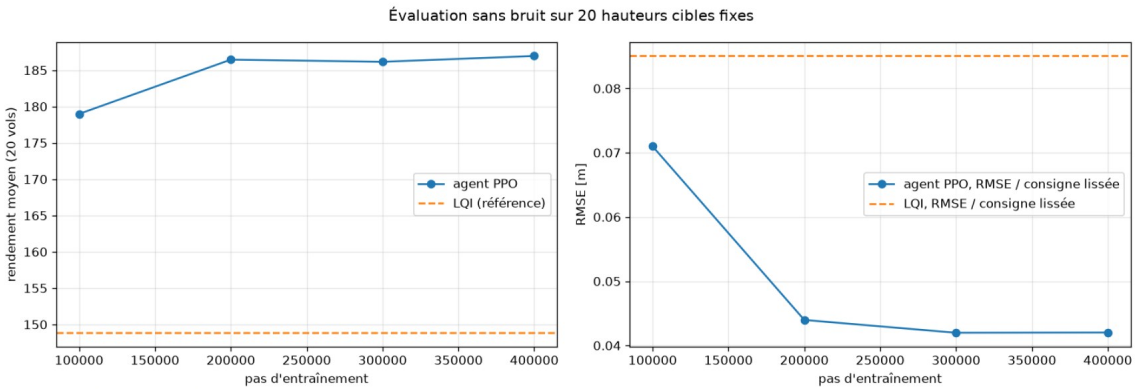
Reward terms vs tracking error: two Gaussians + a linear slope, bounded



SAC: saw-tooth, collapses at 90k steps

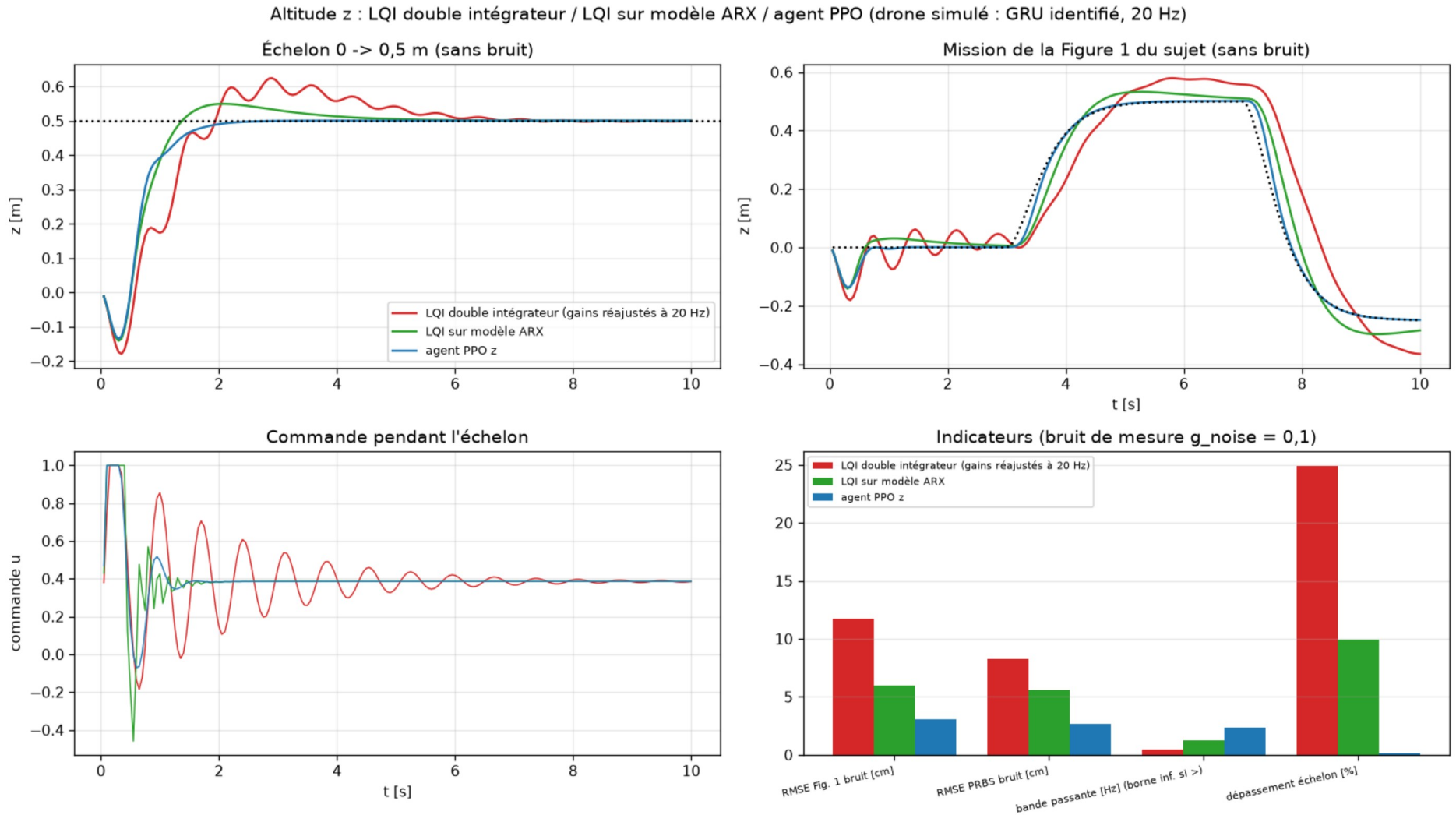


PPO: stable, return 179 → 187

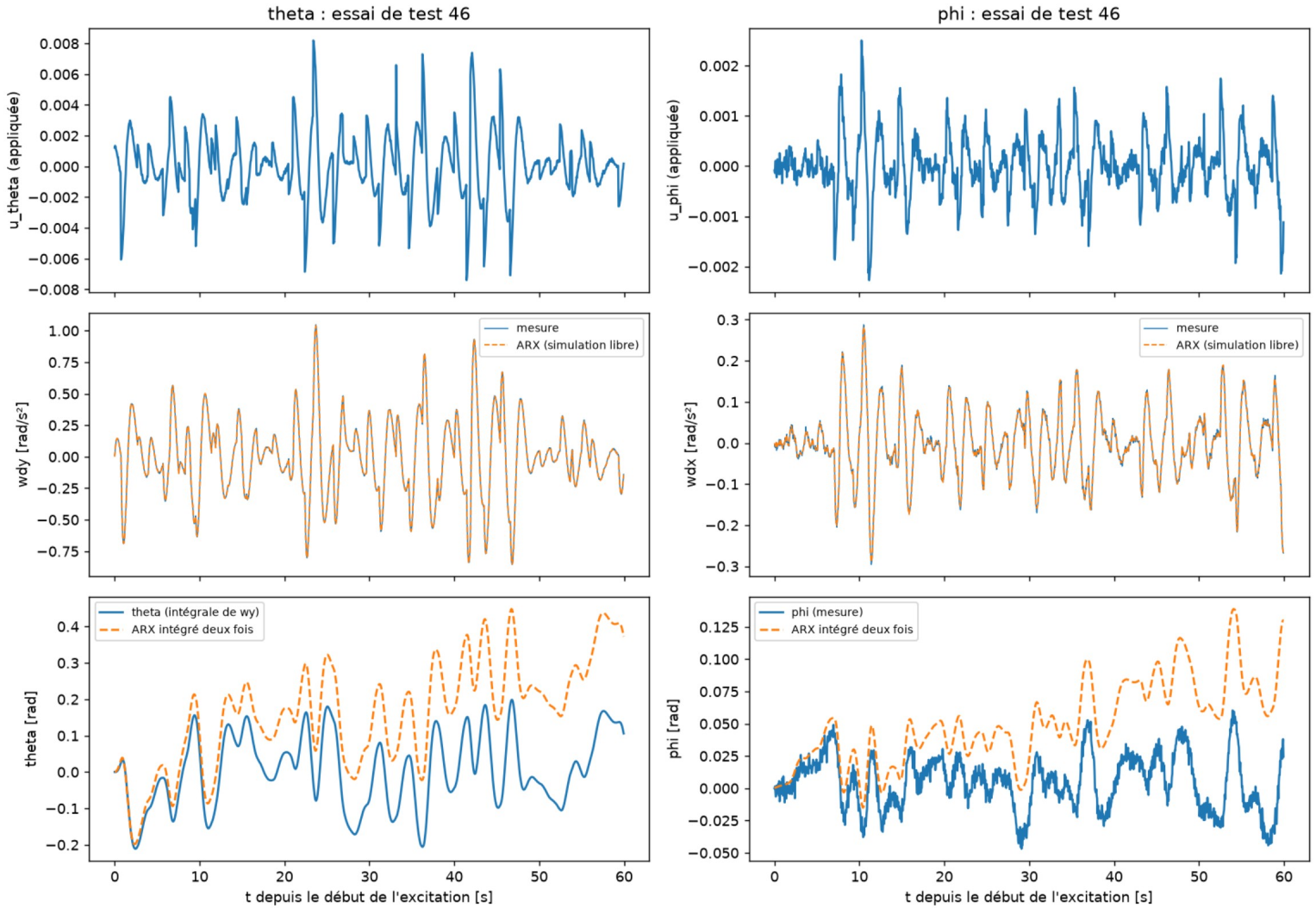


SAC's rewards ranged from −0.01 to −40,000 and its critic loss spiked; PPO with 16 drones, output started at hover (u = 0.39)

The altitude agent tracks with 2.7 cm error, against 5.6 cm for the LQ on the ARX model and 8.3 cm for our LQI, with almost no overshoot.



From 150 one-minute flights we fit a linear ARX model of the tilt: as precise as the sensor noise, and it captures the 0.2 s motor lag.



Free-run test flight: the ARX(2,6) only receives the command and reproduces angular acceleration, rate and angle (measured vs model)

A fast loop holds the tilt, a slower loop turns the position error into a tilt; only the front and rear motors move the drone forward.

x_{ref}, v_{ref}
from the car

→

OUTER LOOP · SLOW
x agent (or PD)
$\theta_{ref} = 0.2 \cdot a$, $\omega_n \approx 1.5$ rad/s

θ_{ref} →

INNER LOOP · FAST
LQ on the ARX model
0.30 s rise, motor lag included

u_θ →

MIXER
$u_1 = u_z - u_\theta$
$u_3 = u_z + u_\theta$

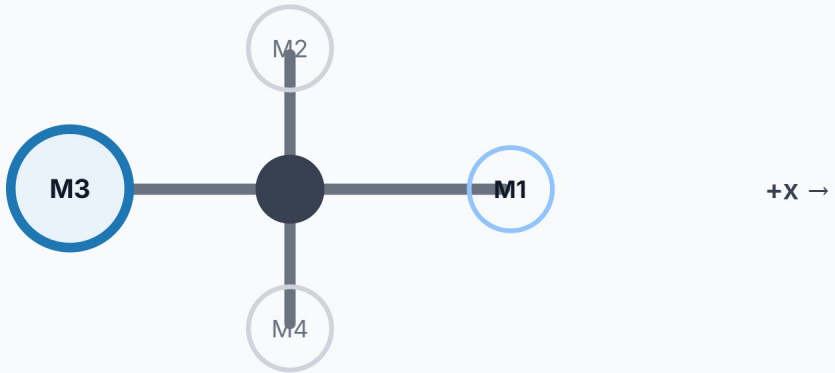
→

DRONE
$\theta \rightarrow a_x \approx g \cdot \theta$
$\rightarrow v_x \rightarrow x$

θ, ω_y fed back every 50 ms

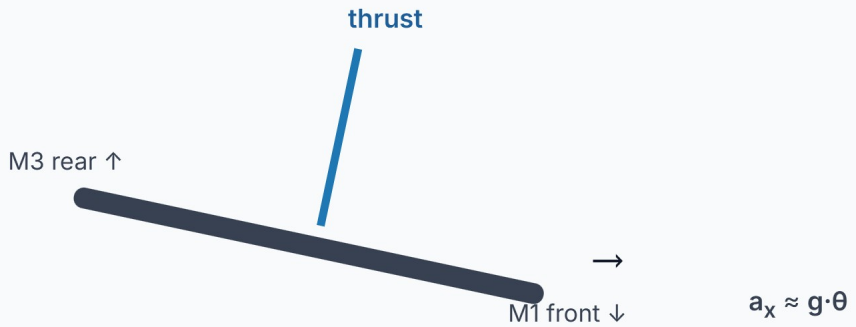
x, v_x fed back: the outer loop is 3–5× slower

Top view, "+" layout



rear M3: + u_θ · front M1: − u_θ · M2, M4: unchanged (on the pitch axis)

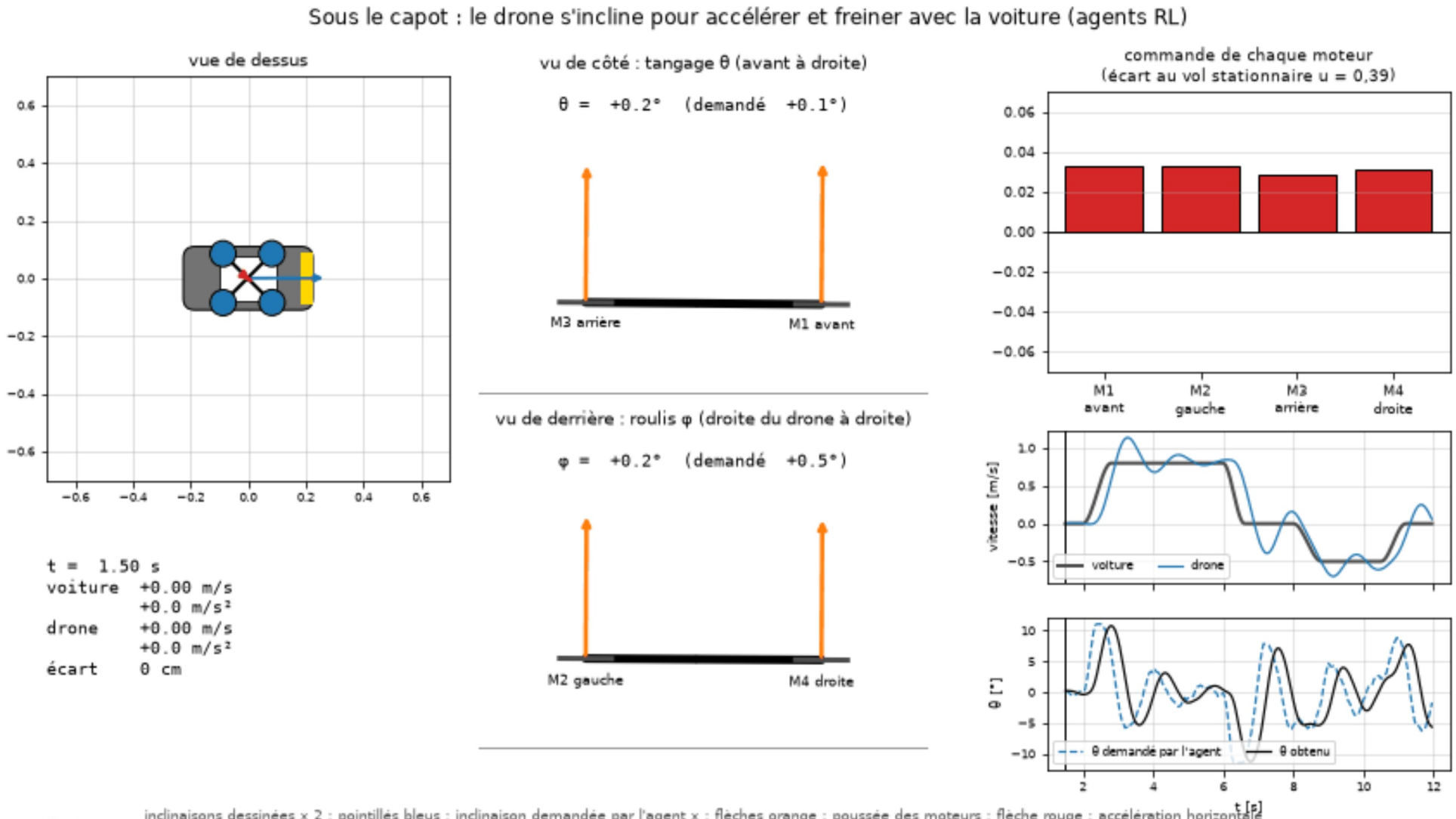
Side view



tilt $\theta \leq 0.2$ rad, so at most ≈ 2 m/s² of acceleration

Same structure for y with φ, motors M2 and M4

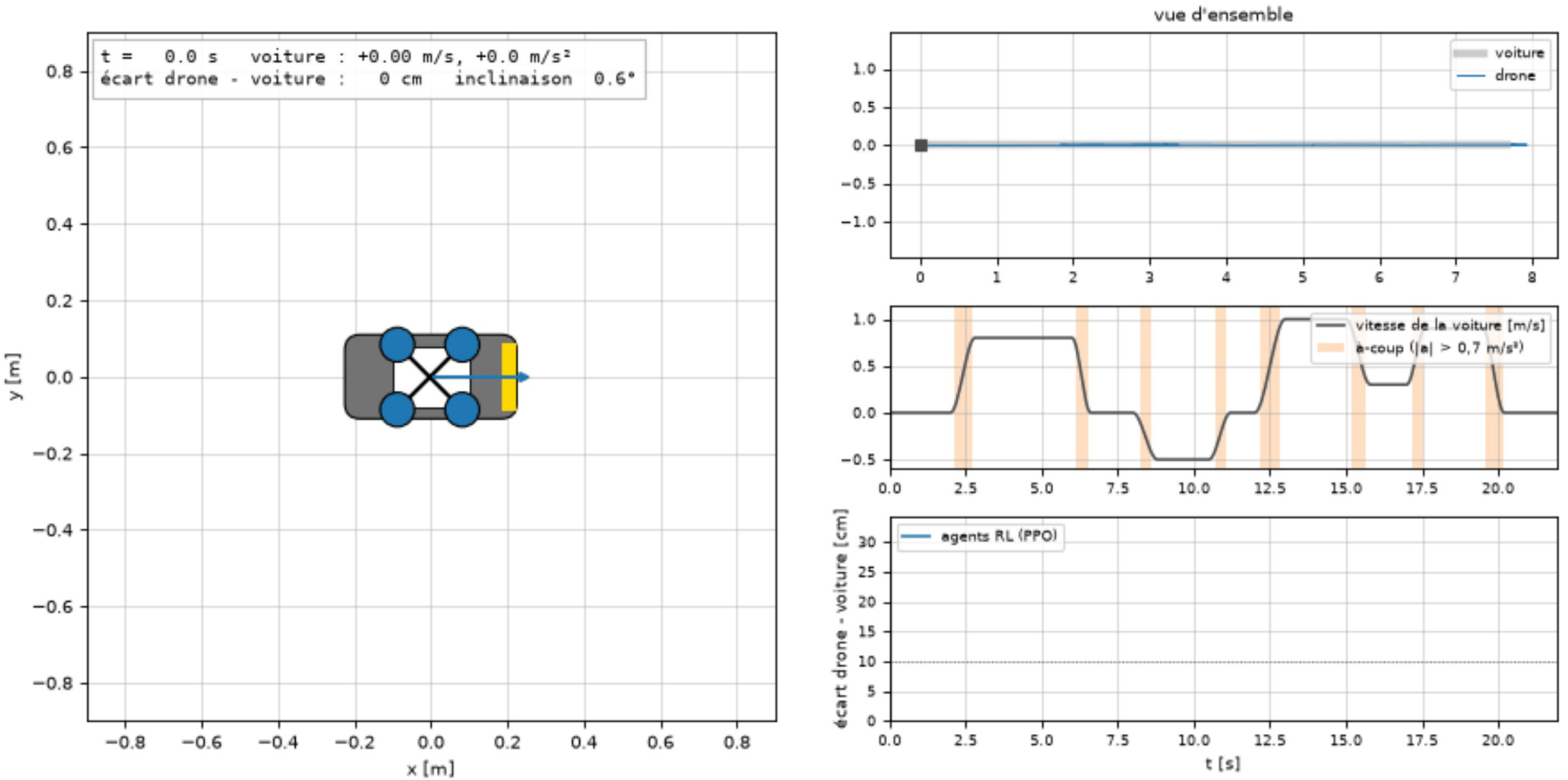
The x agent only asks for a tilt; an LQ designed on the ARX model tracks it, motor delay included.



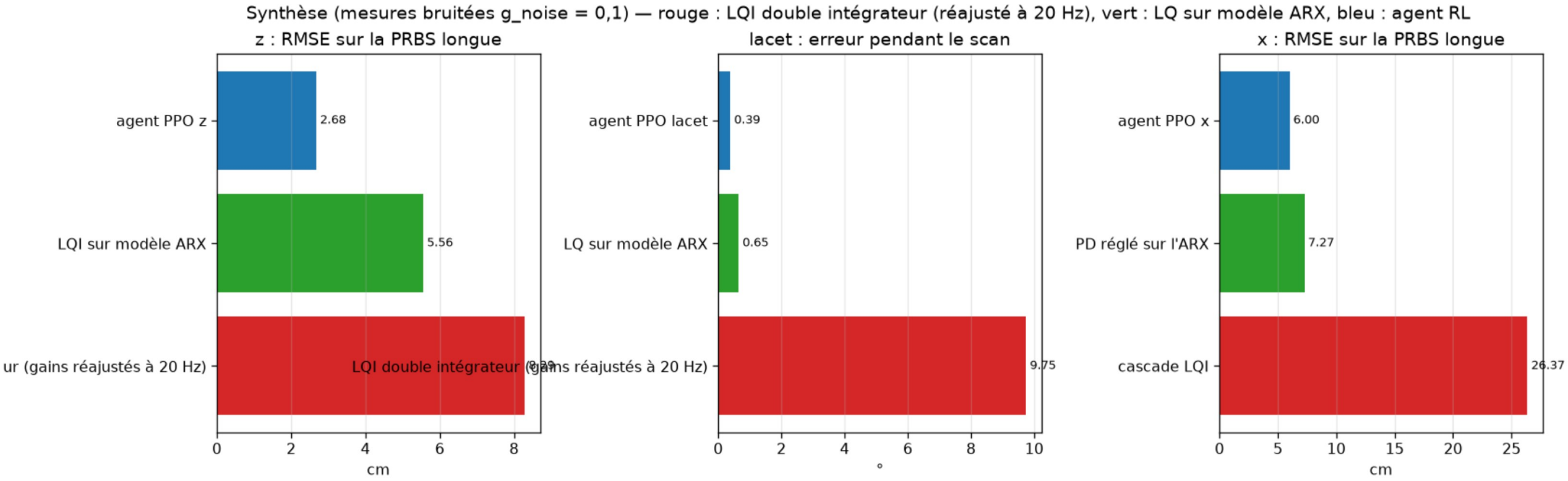
Start, brake, reverse · tilt drawn ×2 · four motor thrusts · θ asked by the agent vs θ obtained

Retrained on saturating targets, one x agent drives both axes and keeps up with a car braking at the drone's 2 m/s² limit.

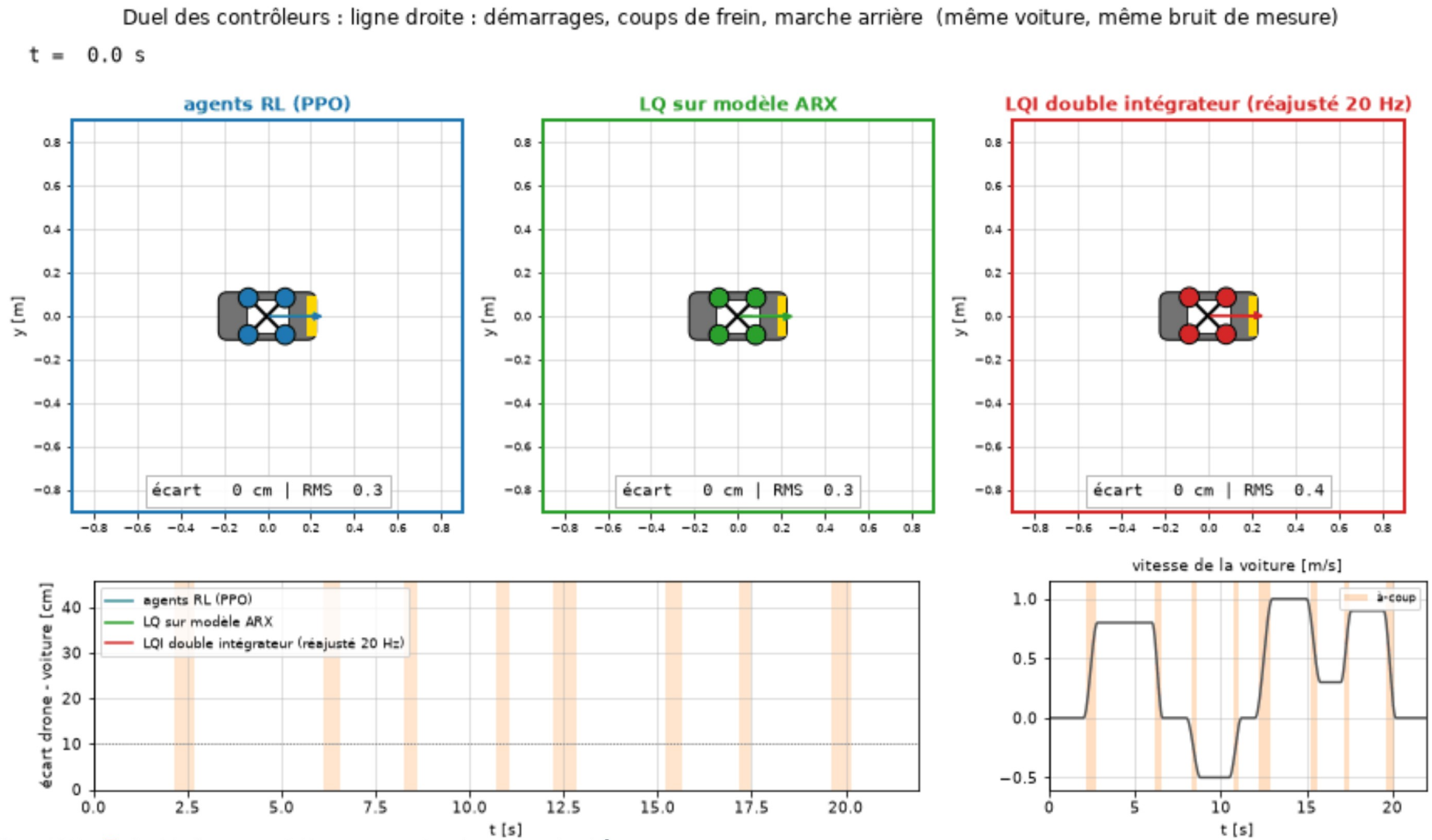
Le drone reste au-dessus de la voiture : ligne droite : démarrages, coups de frein, marche arrière



Modelling the motors brings the first gain; learning brings the second: the RL agents halve the best LQ's altitude error.

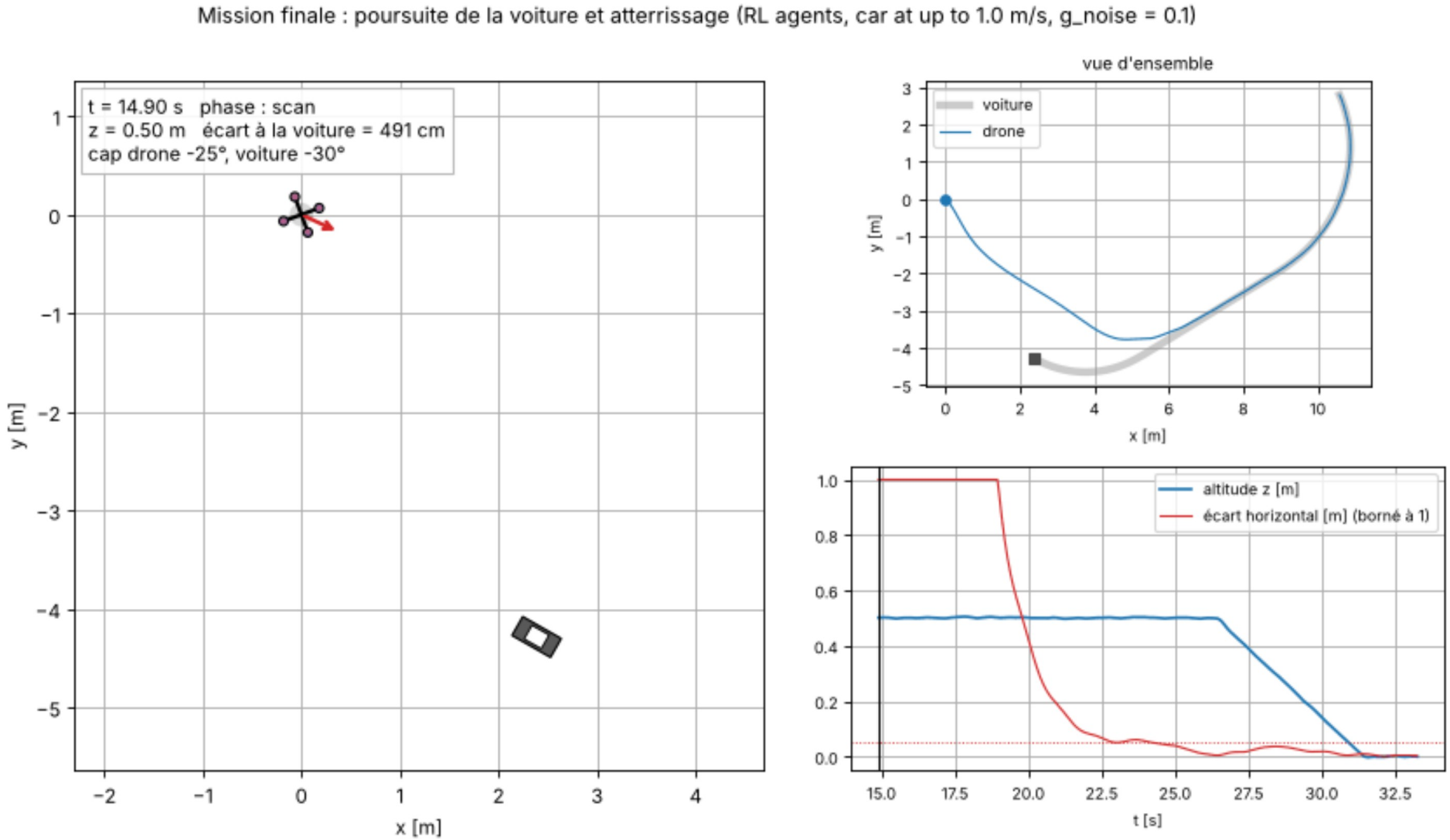


Same car, same noise: the RL agents stay closest, 12.1 cm RMS against 15.1 cm and 16.3 cm.

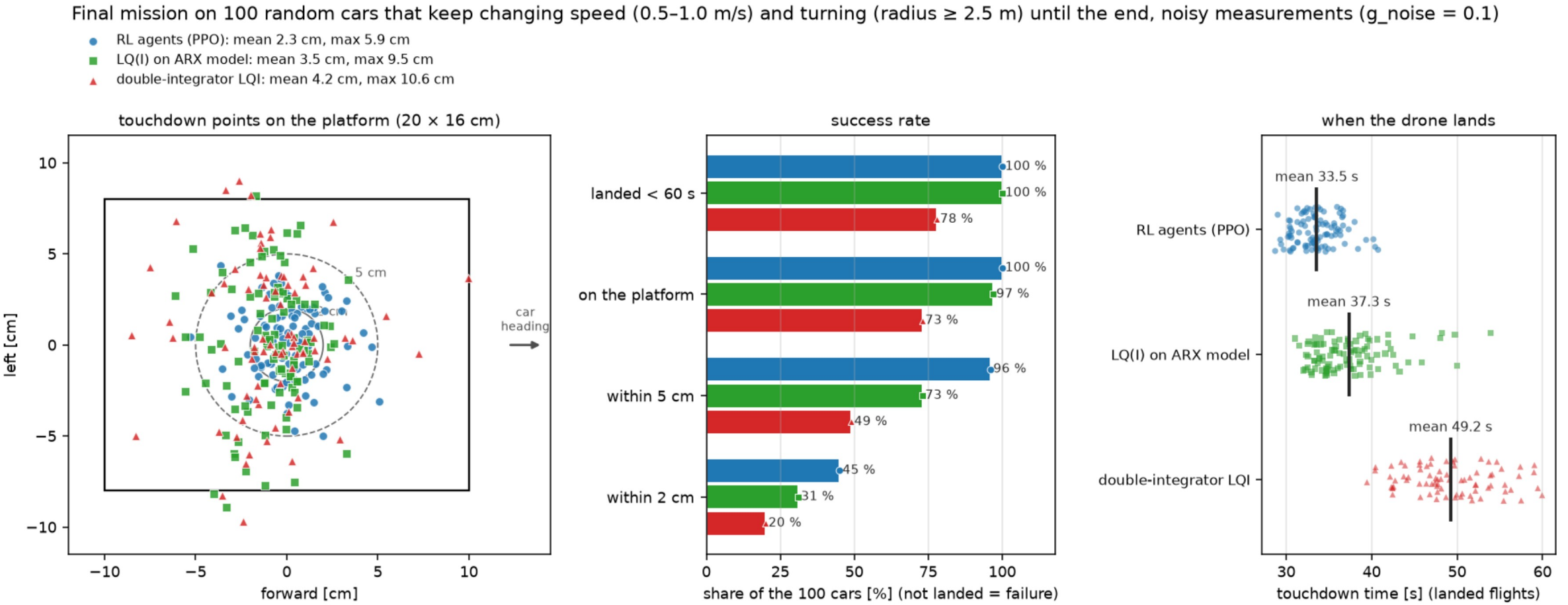


RL agents · LQ(I) on ARX · double-integrator LQI · car accelerations up to 2 m/s²

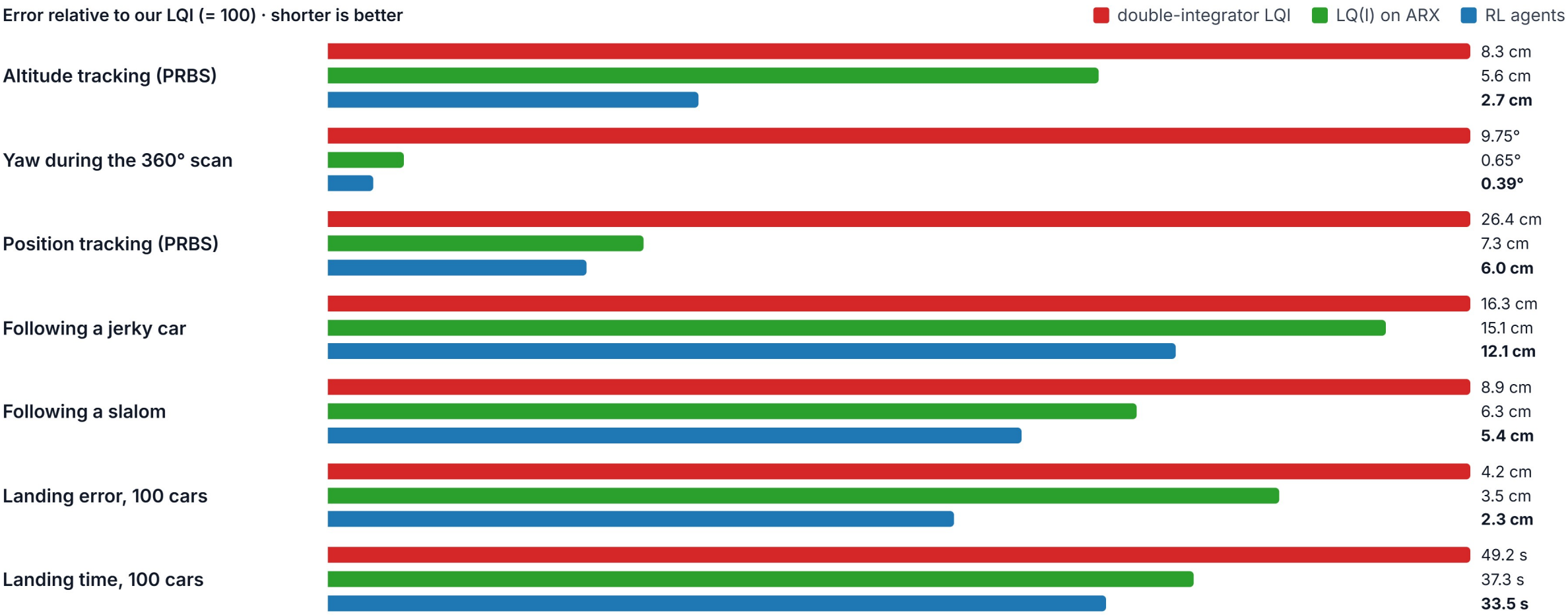
On a car turning at up to 1 m/s, the agents catch up, align and land 1 cm from the centre.



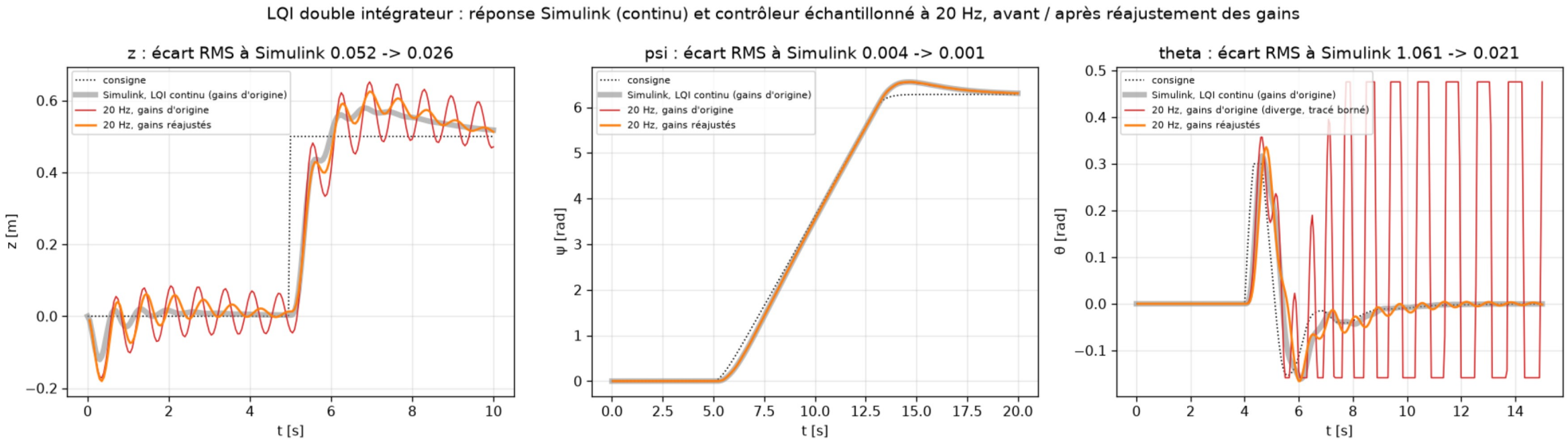
On 100 cars that never stop manoeuvring, the RL agents land every time, 96 % within 5 cm, 15.7 s before the LQI.



On every task, from tracking to landing, the RL agents have the smallest error of the three controller families.

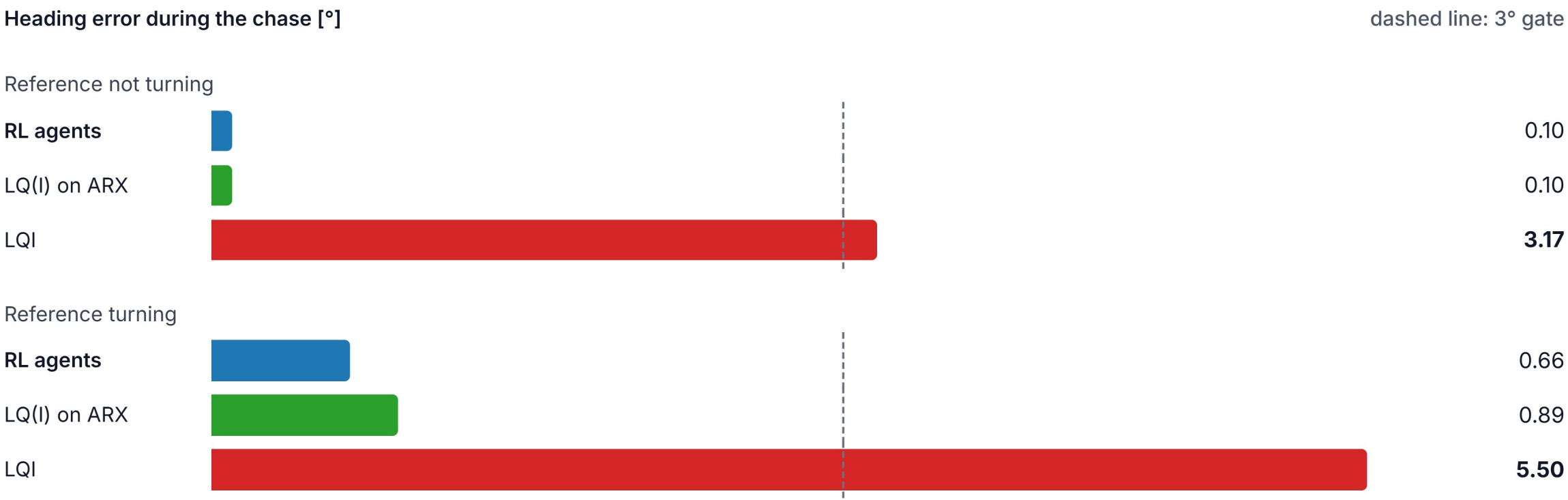


The LQI keeps its control law; its gains are fit so the 20 Hz response matches its own continuous Simulink response.



Continuous Simulink · 20 Hz with the original gains (θ diverges) · 20 Hz with the refit gains

The LQI's yaw lags a turning car by 0.51 s × ω, so its heading sits above the 3° gate that opens the descent.



Measured on the 13 reference cars, Python simulator, g_noise = 0.1

Every design decision was kept only if its impact could be measured.

2	**Integral state on z**	static error removed (0.9 m → 1 m)
2	**Refit LQI gains for 20 Hz**	z 5.2 → 2.6 cm, θ diverging → 0.021 rad
3	**Drop whole corrupted trials**	2996 trials, GRU at the noise floor
3	**GRU for z, ARX for yaw and attitude**	ARX of z alone → 10 % overshoot
4	**Bounded 5-term reward**	overshoot 0.009 vs 0.082 m (SAC)
4	**Conditional integral**	final error 0.8 mm vs 3.9 cm
4	**PPO with 16 envs**	5,248 vs 67 transitions/s
4	**Output bias = hover 0.39**	160 m failure → convergence
5	**Agent → θref, LQ on ARX inner loop**	attitude overshoot 3 % vs 32 %
5	**Faster inner LQ (R = 200)**	unstable → stable, 0.30 s rise
6	**Retrain x on saturations**	car overshoot 0.60 → 0.02 m
6	**Same agent on y**	no retraining, y PRBS 0.060 m
6	**Rotate errors by ψ**	900 m divergence → 0.090 m
7	**Velocity feedforward in the PD**	lag 36 → 0.1 cm
9	**Supervisor decided on measurements**	Simulink vs Python ≤ 7 mm
9	**Keep τ = 1 s**	shorter τ: no faster, more overshoot
9	**Subtract the agent bias a(0)**	chase 8.7 → 8.4 s
9	**Point the heading at the car**	touchdown heading −40° → < 0.3°
9	**Fair alignment criterion**	agent better on every robustness metric
10	**Cars manoeuvre until the end**	“LQI most precise” artefact removed

Left: slide where the decision appears · reasons in each slide's speaker notes